Further results from testing the South African sardine stock assessment model

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Further results from the stock assessment for the revised stock structure hypothesis for South African sardine are presented, exploring some questions raised in de Moor (2024b).

Keywords: movement, population dynamics model, recruitment, sardine, South Africa

Background

This document presents further results from alternative assumptions for the stock assessment given the revised stock structure hypothesis for South African sardine (de Moor 2024a). These alternatives are aimed at contributing to discussion on the amount of WTS spawning / recruitment off the west coast and the associated timing of WTS spawning (de Moor 2024b). In summary, the 'concern' is (i) WTS peak recruitment (off both the south and west coasts) is estimated by S_0 to be after October, in contrast the winter-spring peak spawning period off the south coast as indicated by GSI analyses, and (ii) if WTS spawning / recruitment off the west coast is substantial and should be earlier than that of CTS off the west coast, why was this high winter/spring spawning not evident in the GSI analyses of data collected off the west coast which showed spring-summer peak spawning.

Method

The model against which comparisons are made is S_0 from de Moor (2024b) which does not include parasite prevalence-at-length data in the likelihood and has a time invariant growth curve, i.e. $\eta_{c,y}^{t} = 0$.

First, WTS off the west coast are assumed to have a peak recruitment reflecting that of the west coast (CTS), rather than WTS off the south coast:

S_{13}: S_0 with $t_{0,w}^{WTS} = t_{0,w}^{CTS}$, $t_{0,s}^{CTS} = t_{0,s}^{WTS}$ instead of $t_{0,w}^{WTS} = t_{0,s}^{WTS}$, $t_{0,s}^{CTS} = t_{0,w}^{CTS}$

S_{13*}: S_{13} without the constraint of $L_{w,\infty} \leq L_{s,\infty}$

Alternatives restricting WTS recruitment off the west coast:

S_{14}: S_0 with $R_{c,y}^{WTS}/10 \sim U(0,5)$ instead of $R_{c,y}^{WTS}/10 \sim U(0,50)$, with $p_y \sim U(0,1)$[1]

S_{15}: S_{13} with $R_{c,y}^{WTS}/10 \sim U(0,5)$ instead of $R_{c,y}^{WTS}/10 \sim U(0,50)$, with $p_y \sim U(0,1)$[2]

An alternative where WTS recruitment to both coasts is estimated, and then subsequently split by coast (this split therefore incorporates both (south-)west coast spawning and passive movement):

S_{16}: S_0 with $R_{c,y}^{WTS} = q_y \times R_y^{WTS}$, and no p_y

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[1] An alternative with $p_y \sim N(0.325, 0.165^2)$ did not fit the data well.

[2] An alternative with $p_y \sim N(0.325, 0.165^2)$ did not fit the data well.

S_{17}: S_{13} with $R_{c,y}^{WTS} = q_y \times R_y^{WTS}$, and no p_y

Alternatives allowing for annual variability in the juvenile natural mortality rate of WTS off the west coast (potentially reflecting that in some years the environment off the west coast is better suited to WTS):

S_{18}: S_0 with $M_{s,y,0}^{WTS} = \left(M_{w,y,0}^{CTS} + M_{juv}^{diff}\right)e^{\varepsilon_y}$, with $\varepsilon_y \sim N(0, \sigma^2)$ and $\sigma \sim U(0.2, 0.5)$

S_{19}: S_{14} with $M_{s,y,0}^{WTS} = \left(M_{w,y,0}^{CTS} + M_{juv}^{diff}\right)e^{\varepsilon_y}$, with $\varepsilon_y \sim N(0, \sigma^2)$ and $\sigma \sim U(0.2, 0.5)$

To investigate the impact of greater active movement of WTS at younger ages:

S_{20}: S_0 with $move_{y,1}^{WTS} = 1$, i.e. all WTS move to the south coast by the time they are 2 years old

S_{21}: S_0 with $move_{y,0}^{WTS} = 1$, i.e. all WTS move to the south coast by the time they are 1 year old

S_{22}: S_{15} with $move_{y,1}^{WTS} = 1$, i.e. all WTS move to the south coast by the time they are 2 years old

Results and Discussion

Assuming the peak recruitment of WTS off the west coast is the same as that of CTS off the west coast (S_{13}, S_{15}) results in a better fit to the south coast survey length frequency data (Table 1, Figure 4a), and the timing of peak recruitment is estimated to be around early August off the south coast and late November off the west coast (Figure 9a). This aligns more closely with that expected *a priori* (van der Lingen and McGrath 2017). An additional run (S13*) without the constraint of $L_{w,\infty} \leq L_{s,\infty}$ does result in $L_{w,\infty} > L_{s,\infty}$ (Figure 9c).

This alternative (of assuming peak recruitment of WTS off the west coast is the same as that of CTS off the west coast) applies the 'WTS west coast growth curve' to all WTS off the west coast, regardless of origin. If this alternative is pursued, additional work may be required to track WTS originating from spawning off the south coast separately from those originating from spawning off the west coast, as the former would be expected a priori to have an earlier peak recruitment corresponding with that of WTS off the south coast.

If recruitment of WTS off the west coast is restricted to a maximum of 50 billion recruits, and the passive movement has an uninformative prior distribution (S_{14}, S_{15}) the WTS recruitment off the south coast is estimated to be substantially higher in many years (Figure 8a), with a widely varying proportion of that recruitment moving to the west coast annually (Figure 7). The resultant recruitment off the west and south coasts at the time of the recruit survey is similar to S_0 and the fit to the data is thus very similar (slightly better). While this difference has minimal impact on the fit to the data, the resulting stock-recruitment scatter plots differ substantially (Figure 10a), which will impact how future recruitment to these stocks is generated in the upcoming Management Strategy Evaluation. These results may be an indication that an informative prior on WTS recruitment off the west coast should be considered.

There is little difference in terms of fitting to the data for S_{16} and S_{17} compared to S_0, and S_{13}, respectively (Table 1), and the resultant recruitment is also similar (Figure 8b).

Allowing the juvenile natural mortality rate of WTS to vary inter-annually off the west coast (S_{18} and S_{19}) results in a small improvement in the fit to the data compared to S_0 and S_{14}, respectively (Table 1). The estimated annual natural mortality rate is higher than that for S_0 (Figure 11). It might, however, be possible that any gains in survival due to environmental conditions would simultaneously affect both juvenile and adult WTS.

If WTS are forced to move to the south coast before they turn 2 instead of before they turn 3 years of age, the fit to the data is slightly worse; the fit is also worse if WTS are forced to move to the south coast before they turn 1 (Figures 1b to 5b). Figure 9b shows the lower SSB estimated under these scenarios. CTS recruitment is estimated to be higher at the turn of the century under S_{21} to compensate for the lack of WTS biomass off the west coast by November.

References

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Table 1a. The contributions to the objective function from likelihood (equations A29 – A33) and prior components for the models considered in this document together with some key parameter estimates at the joint posterior mode.

		S_0	S_{13}	S_{14}	S_{15}	S_{16}	S_{17}	S_{18}	S_{19}	S_{20}	S_{21}	S_{22}
Objective function		-795.64	-800.74	-761.92[%]	-765.43	-761.21[%]	-765.38	-823.33	-789.88	-805.51	-791.55	-764.83[%]
log likelihood	$-lnL$	-806.50	-813.37	-807.23	-813.20	-806.82	-813.00	-810.73	-807.60	-795.40	-784.44	-812.37
	$-lnL^{Nov}$	41.34	41.08	41.22	41.31	41.49	41.46	41.39	40.84	41.67	47.52	41.42
	$-lnL^{rec}$	28.76	30.51	27.72	29.20	28.13	29.38	28.21	27.32	28.06	27.91	29.12
	$-lnL^{sur\ propl}$	-408.81	-415.86	-407.90	-415.45	-408.74	-415.52	-411.99	-408.05	-406.31	-405.30	-413.40
	$-lnL^{com\ propl}$	-467.78	-469.10	-468.28	-468.26	-467.70	-468.31	-468.33	-467.71	-468.93	-461.69	-469.52
	$-lnL^{prev}$	-	-	-	-	-	-	-	-	-	-	-
log priors	$ln(k_{ac}^{S})$	-0.42	-0.41	-0.51	-0.51	-0.47	-0.59	-0.43	-0.50	-0.59	-0.19	-0.61
	$move_{y,1}$	4.21	4.22	4.16	4.00	4.12	3.94	4.41	4.13	4.11	-	3.89
	$\bar{l}_{1,y}$	40.86	40.92	40.76	41.20	40.80	41.19	41.15	40.72	40.84	41.63	41.22
	p_y	-35.05	-35.15	-	-	-	-	-35.05	-	-35.02	-35.18	78.55
	ϑ_a	0.35	0.37	0.35	0.37	0.35	0.37	0.30	0.36	0.36	0.36	0.38
	S_{50}	0.88	2.66	0.53	2.69	0.79	2.68	4.30	0.39	0.39	0.47	2.64
	$\eta_{c,y}^{t}$	-	-	-	-	-	-	-	-	-	-	-
	k_{ac}^{S}	0.65	0.66	0.67	0.68	0.66	0.70	0.65	0.66	0.70	0.95	0.95
	$k_{w,r}$	0.25	0.24	0.24	0.26	0.25	0.27	0.24	0.24	0.25	0.36	0.36
	$k_{s,r}$	0.24	0.24	0.23	0.25	0.24	0.27	0.24	0.23	0.24	0.35	0.35
	$M_{s,y,0}^{WTS}$	1.27	1.46	1.50*	1.50*	1.38	1.50*	1.50*	1.50*	1.50*	1.49	1.49
	$M_{s,y,1^+}^{WTS}$	1.27	1.35	1.27	1.31	1.26	0.28	1.26	1.30	1.29	1.01	1.01
	M_{inc}^{WTS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

* Upper limit of range constraint.

[%] A positive definite Hessian was not found at this solution (just one run from 'uninformed' starting parameter values).

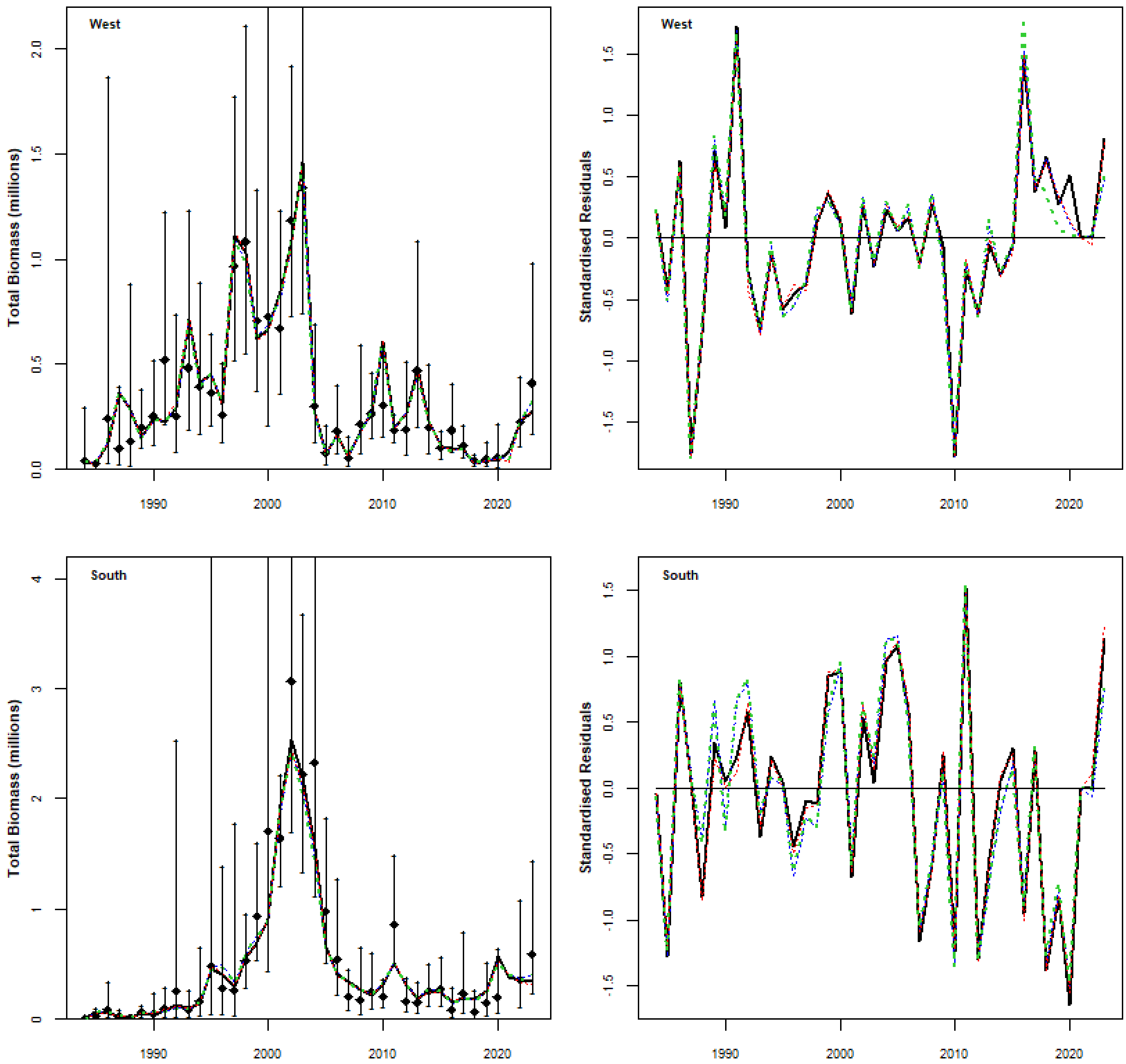


Figure 1a. Acoustic survey estimated and model predicted November sardine total biomasses from 1984 to 2023 for S_0 (black solid line), S_{13} (blue dotted line), S_{14} (red dashed line) and S_{15} (green dashed line). The observed indices are shown with 95% confidence intervals. The standardised residuals (i.e. the residual divided by the corresponding standard deviation, including additional variance where appropriate) from the fits are given in the right side plots.

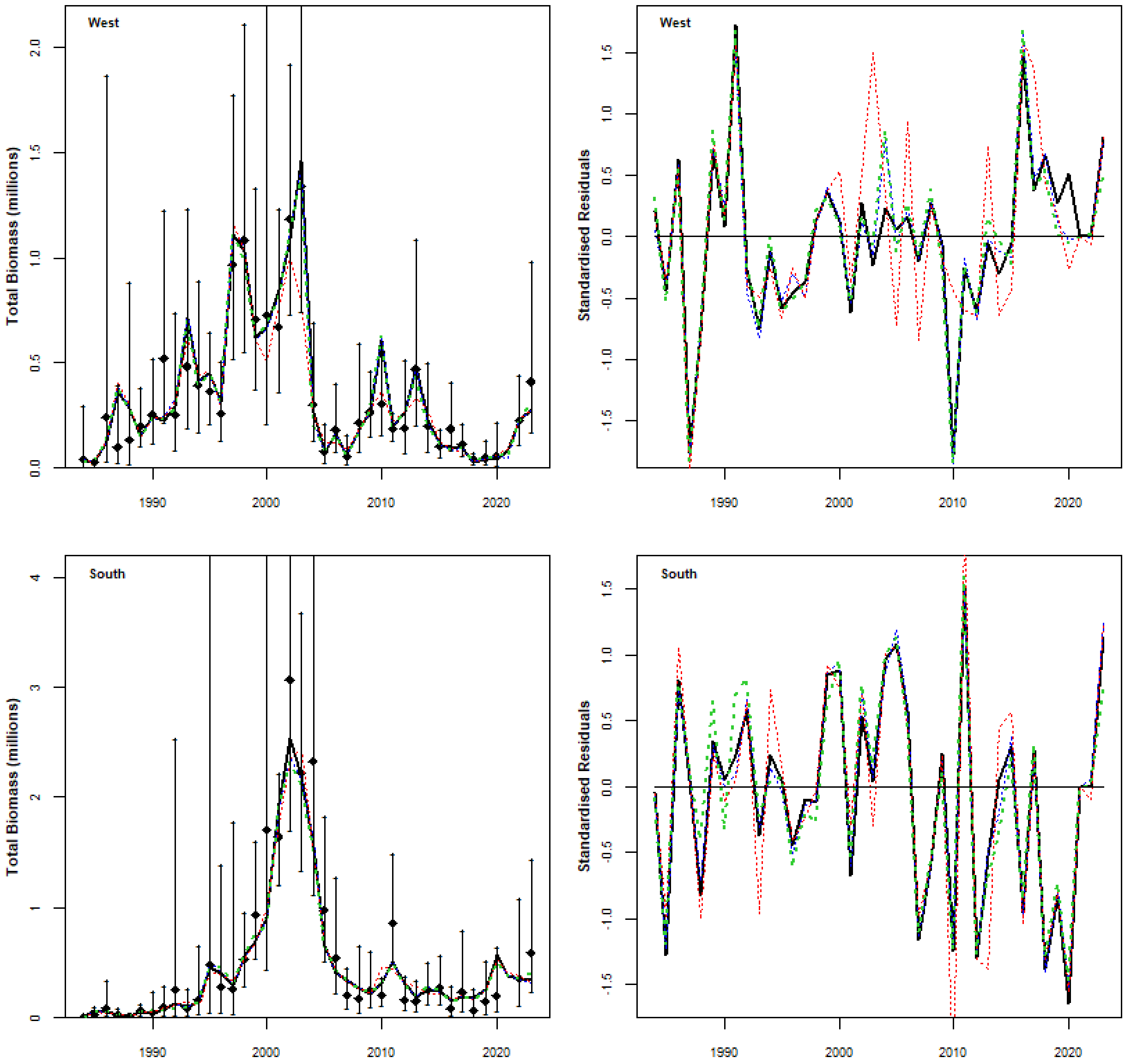


Figure 1b. As for Figure 1a, but for S_0 (black solid line), S_{20} (blue dotted line), S_{21} (red dashed line) and S_{22} (green dashed line).

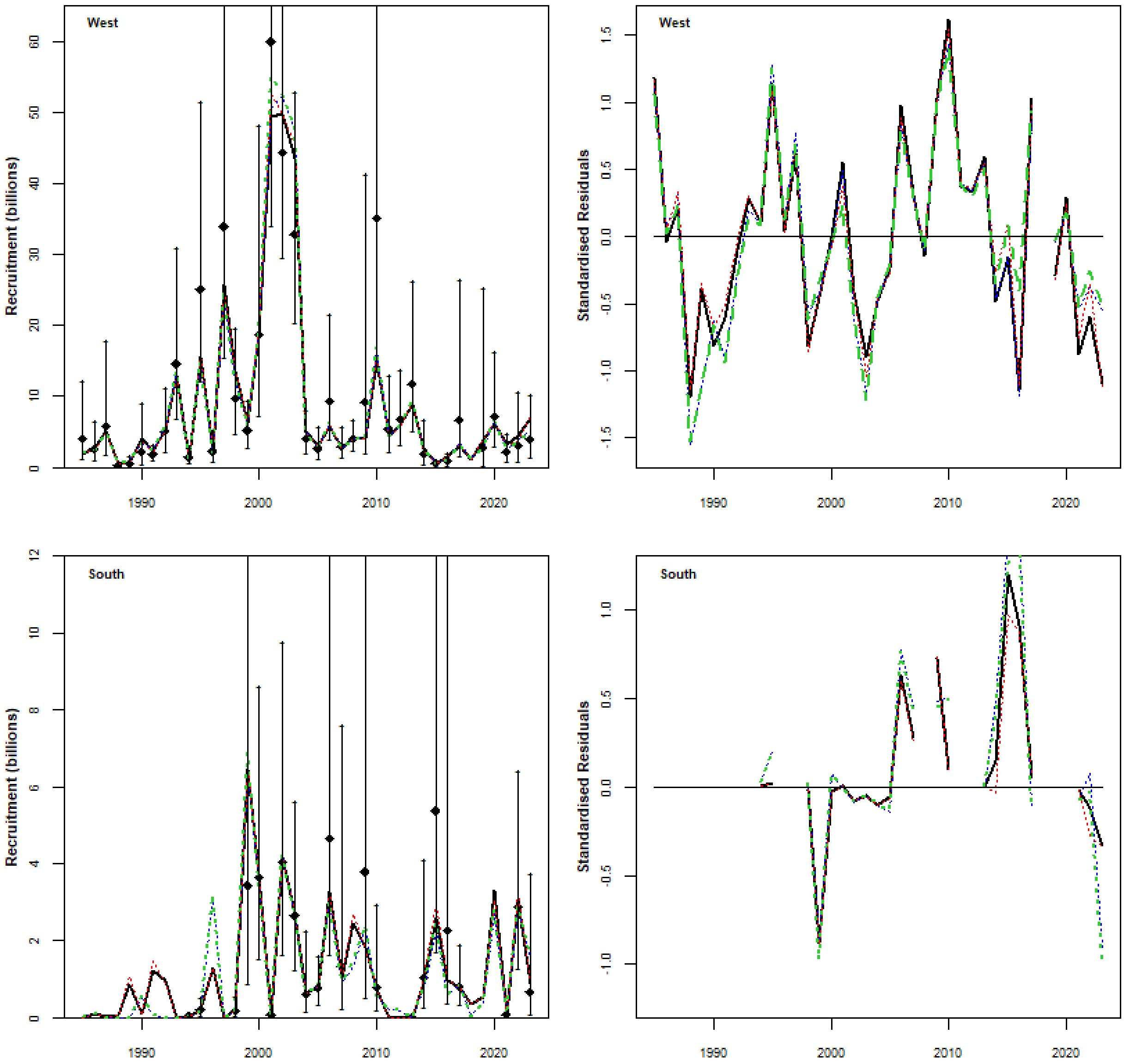


Figure 2a. Acoustic survey estimated and model predicted sardine recruitment numbers from May/June 1985 to 2023 for S_0 (black solid line), S_{13} (blue dotted line), S_{14} (red dashed line) and S_{15} (green dashed line). There was no survey observation in 2018; the model predicted value corresponds to the recruitment predicted at 8th June 2018 which is the average start date of the survey from 2016, 2017 and 2019 surveys. The survey indices are shown with 95% confidence intervals. The standardised residuals from the fit are given in the right side plots.

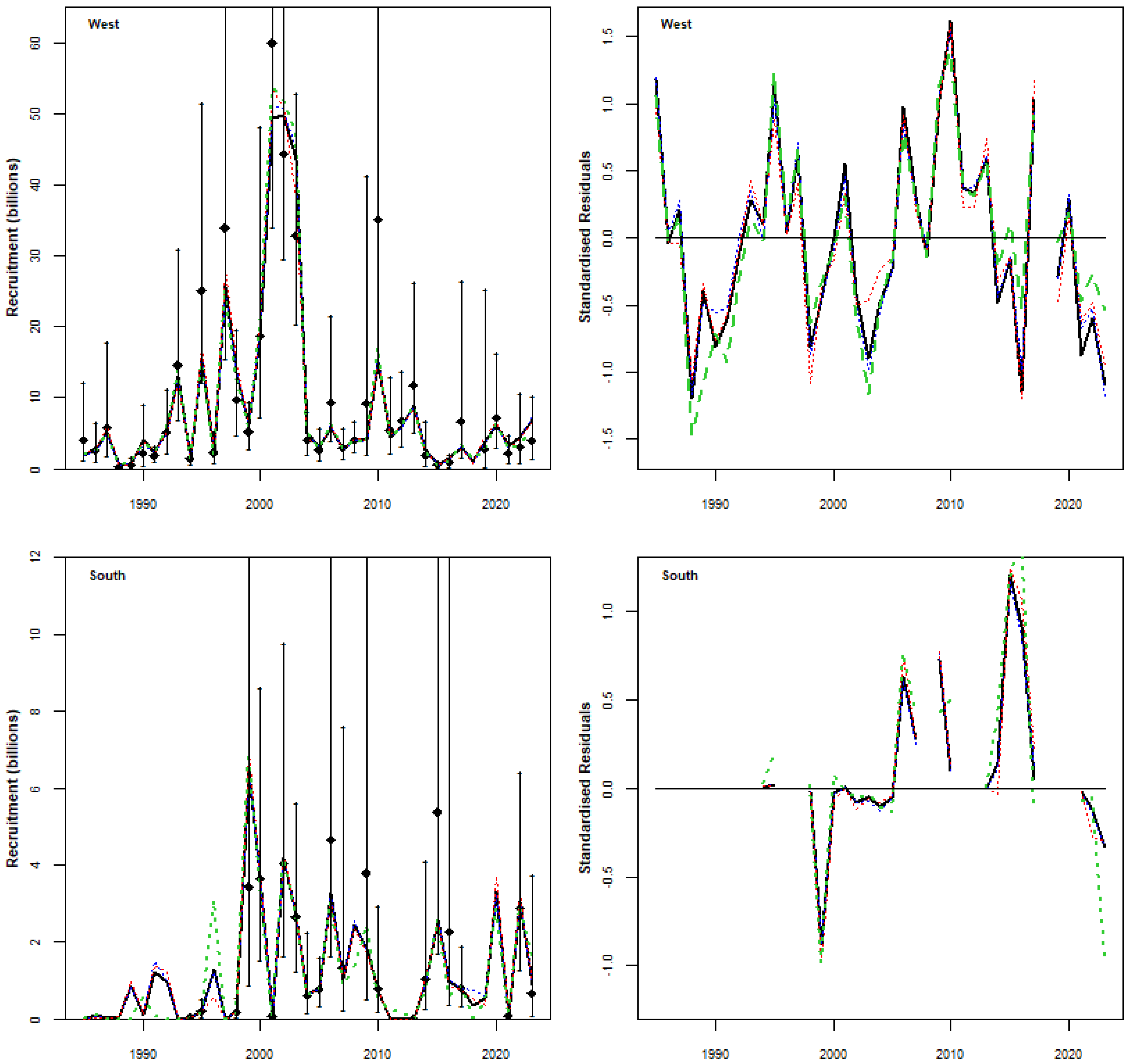


Figure 2b. As for Figure 2a, but for S_0 (black solid line), S_{20} (blue dotted line), S_{21} (red dashed line) and S_{22} (green dashed line).

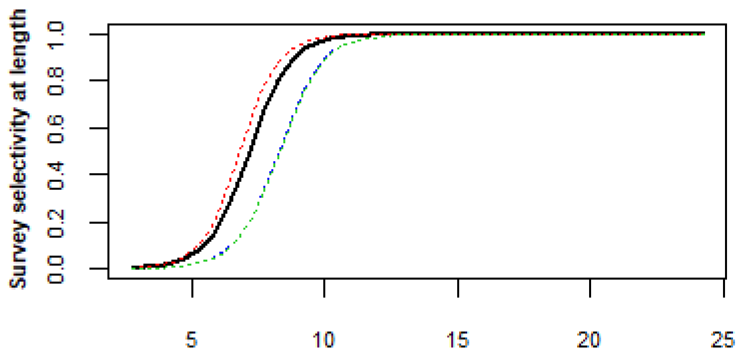


Figure 3a. The model estimated November survey selectivity at length for S_0 (black solid line), S_{13} (blue dotted line), S_{14} (red dotted line) and S_{15} (green dotted line).

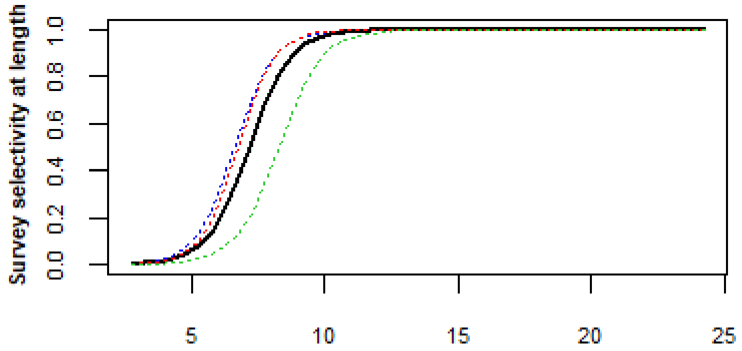


Figure 3b. As for Figure 3a, but for S_0 (black solid line), S_{20} (blue dotted line), S_{21} (red dashed line) and S_{22} (green dashed line).

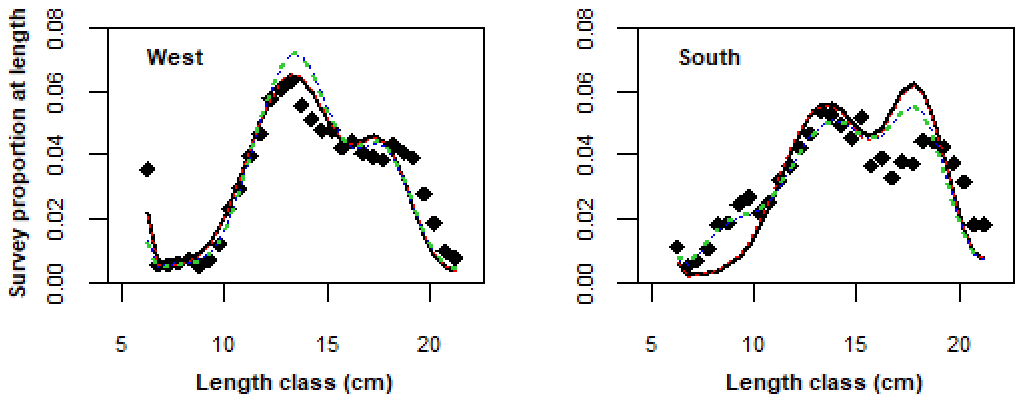


Figure 4a. Average (over all years) model predicted and observed proportions-at-length in the November survey for S_0 (black solid line), S_{13} (blue dotted line), S_{14} (red dashed line) and S_{15} (green dashed line).

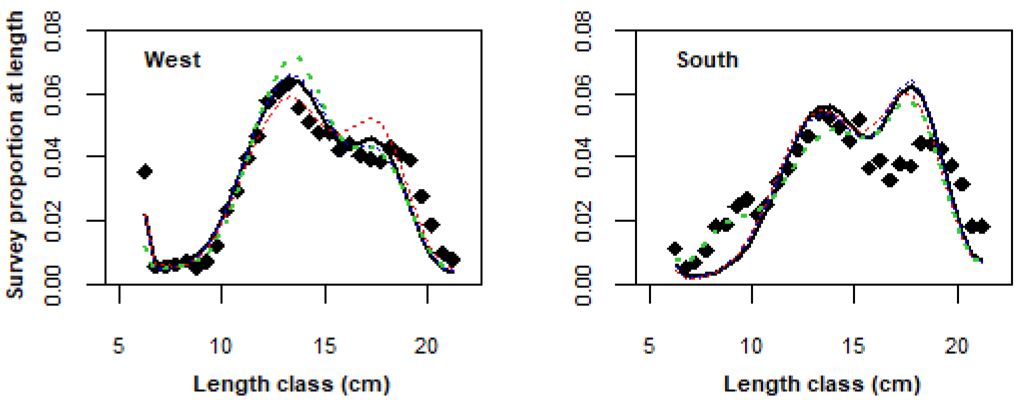


Figure 4b. As for Figure 4a, but for S_0 (black solid line), S_{20} (blue dotted line), S_{21} (red dashed line) and S_{22} (green dashed line).

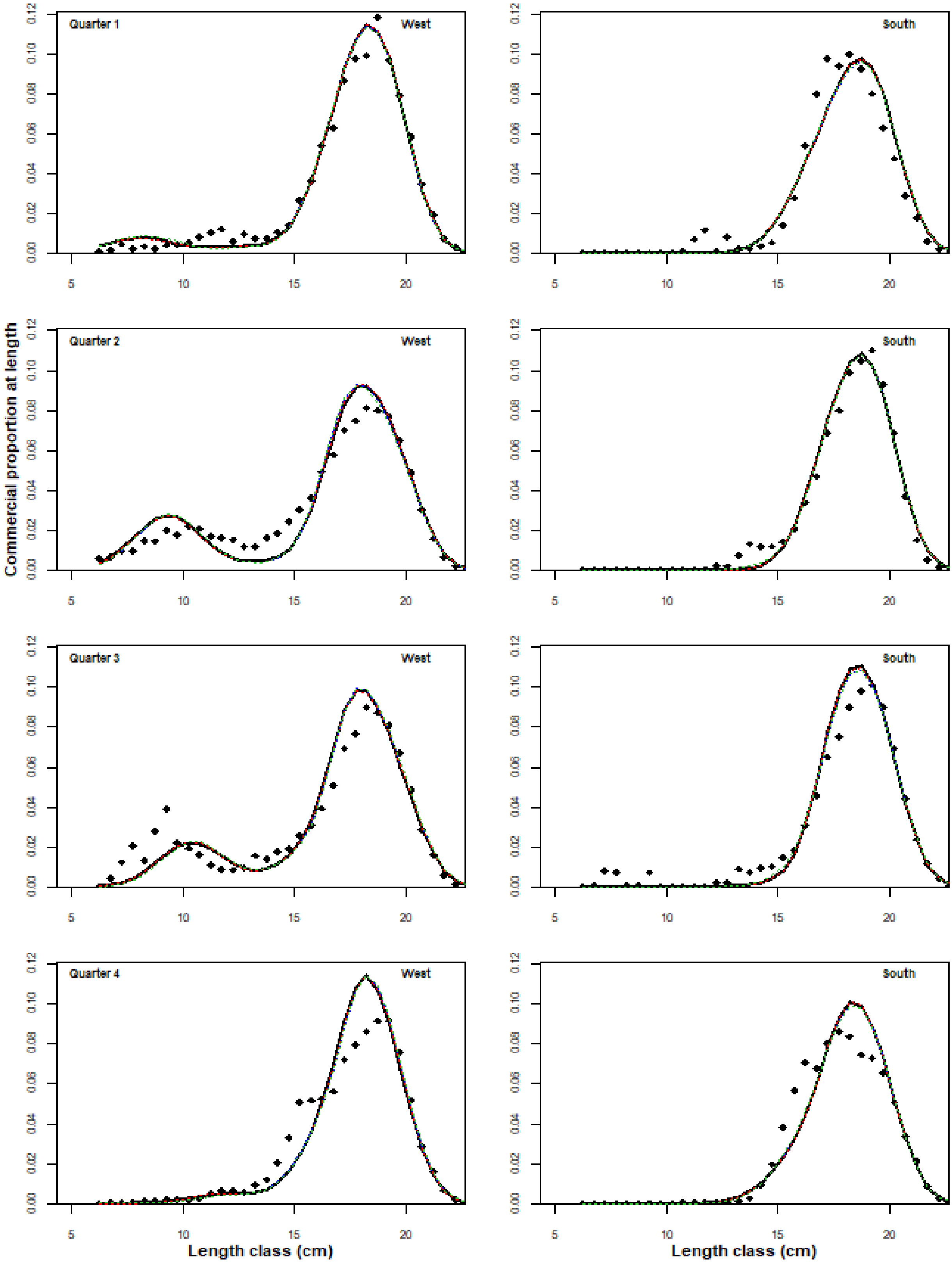


Figure 5a. Average (over all years) quarterly model predicted and observed proportions-at-length in the commercial catch for S_0 (black solid line), S_{13} (blue dotted line), S_{14} (red dashed line) and S_{15} (green dashed line).

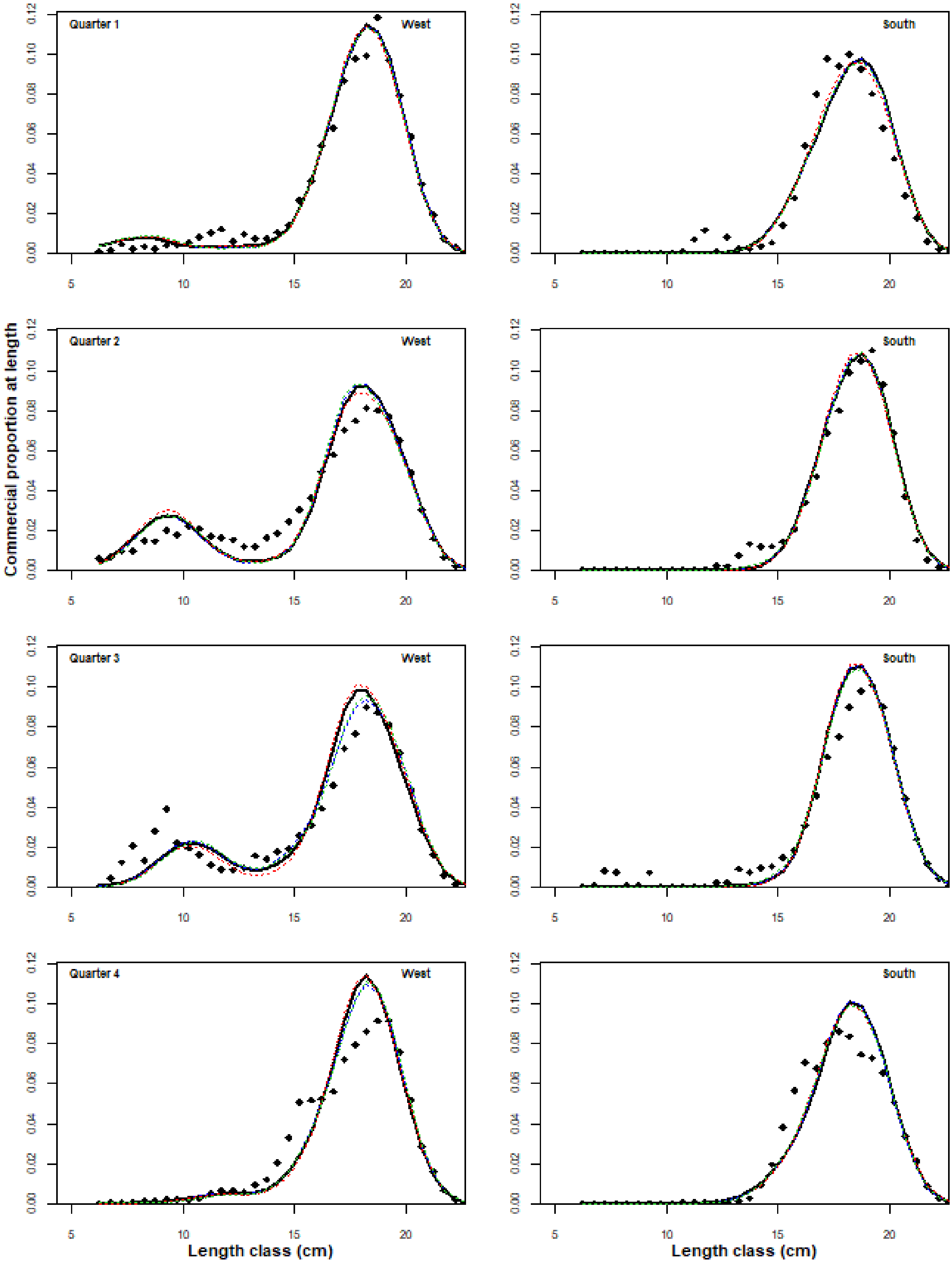


Figure 5b. As for Figure 5a, but for S_0 (black solid line), S_{20} (blue dotted line), S_{21} (red dashed line) and S_{22} (green dashed line).

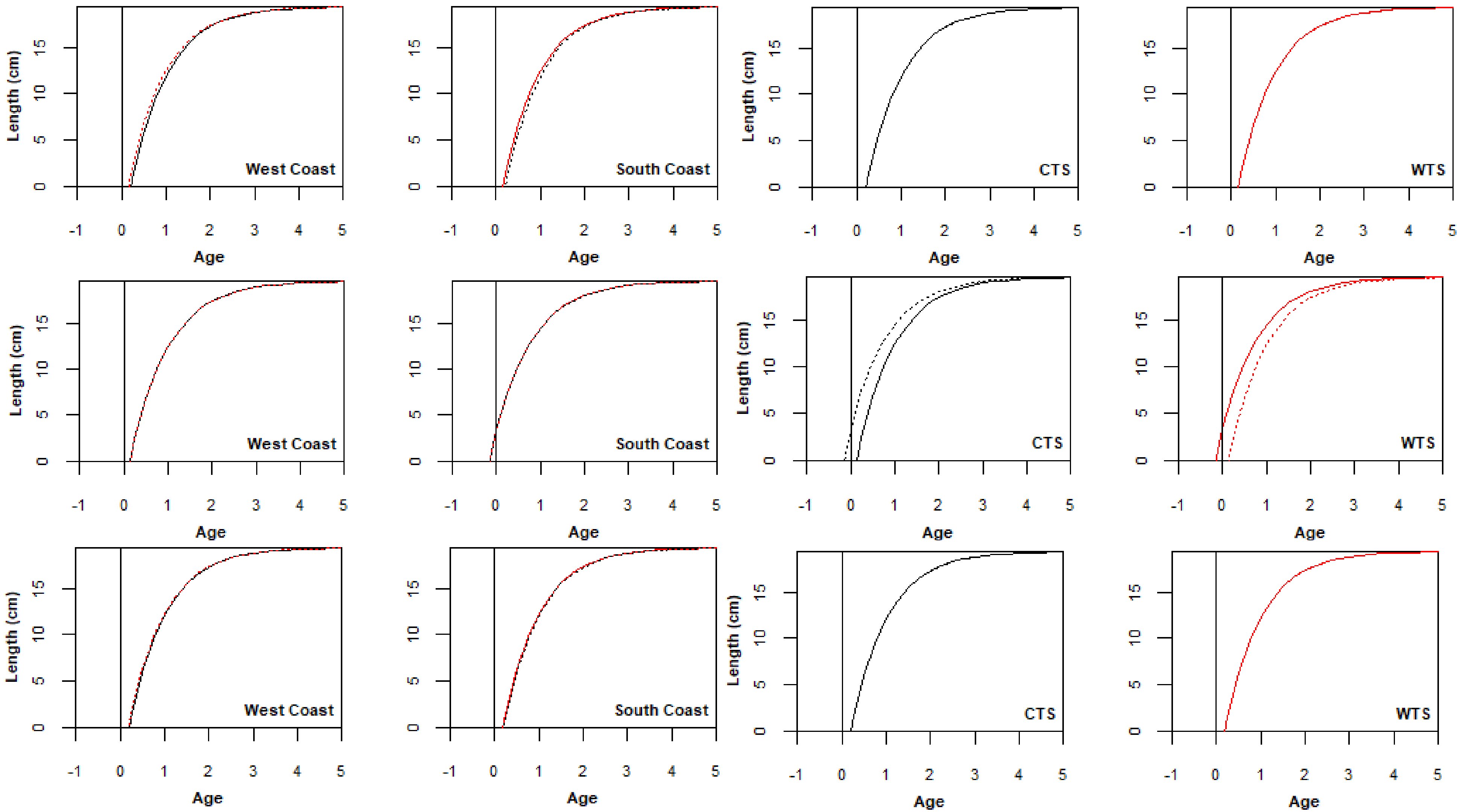


Figure 6a. The von Bertalanffy growth curves for S_0 (top), S_{13} (middle) and S_{14} (lower). (No differences in cohorts assumed). The left-most two plots compare the growth curves between stocks on each coast while the right-most two plots compare the growth curves between coasts for each stock, assuming no movement between coasts. WTS on the South Coast is denoted by a solid red line, and by a dashed red line on the West Coast. CTS on the West Coast is denoted by a solid black line, and by a dashed black line on the South Coast.

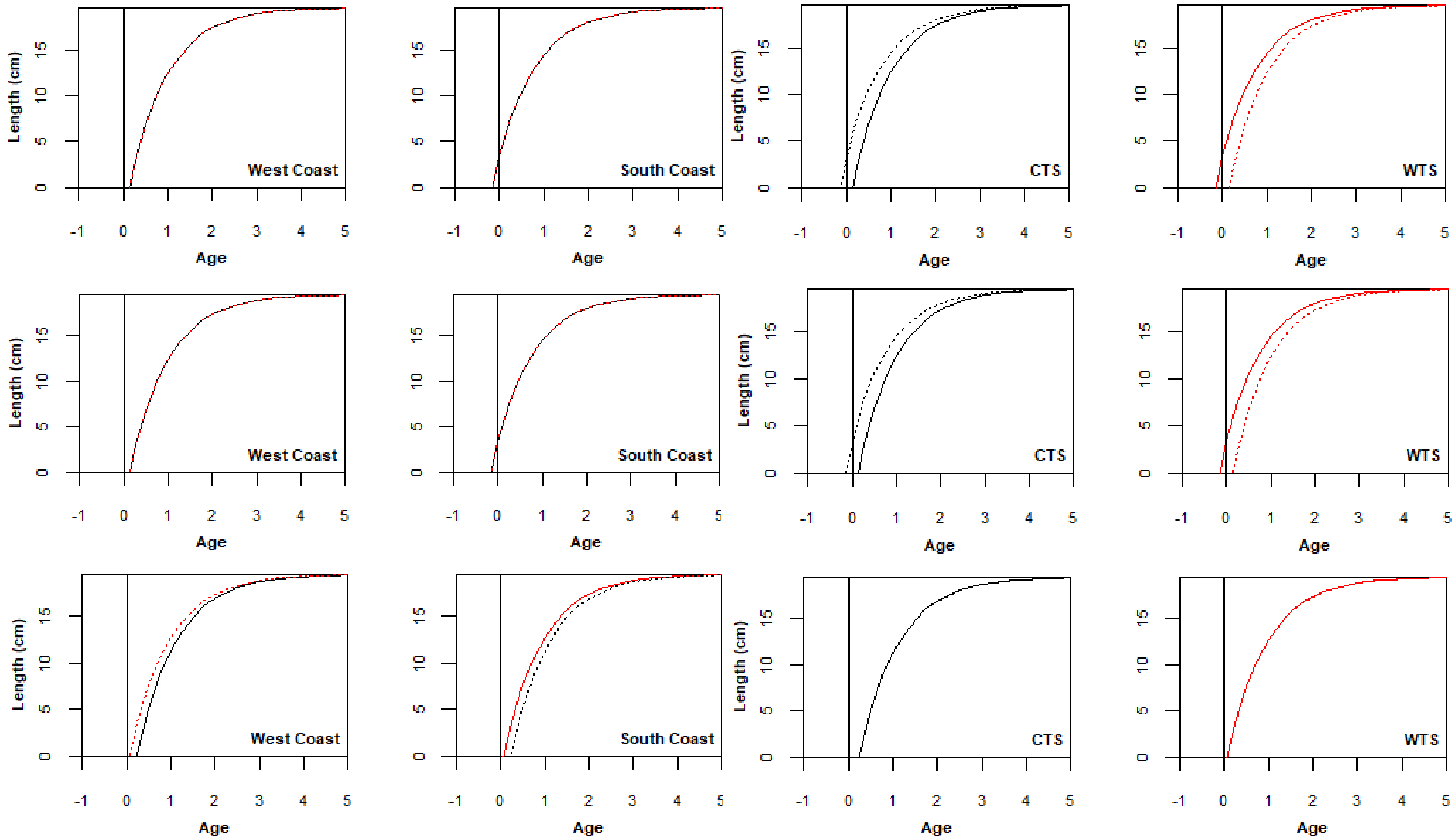


Figure 6b. As for Figure 6a, but for S_{15} (top), S_{17} (middle) and S_{18} (lower).

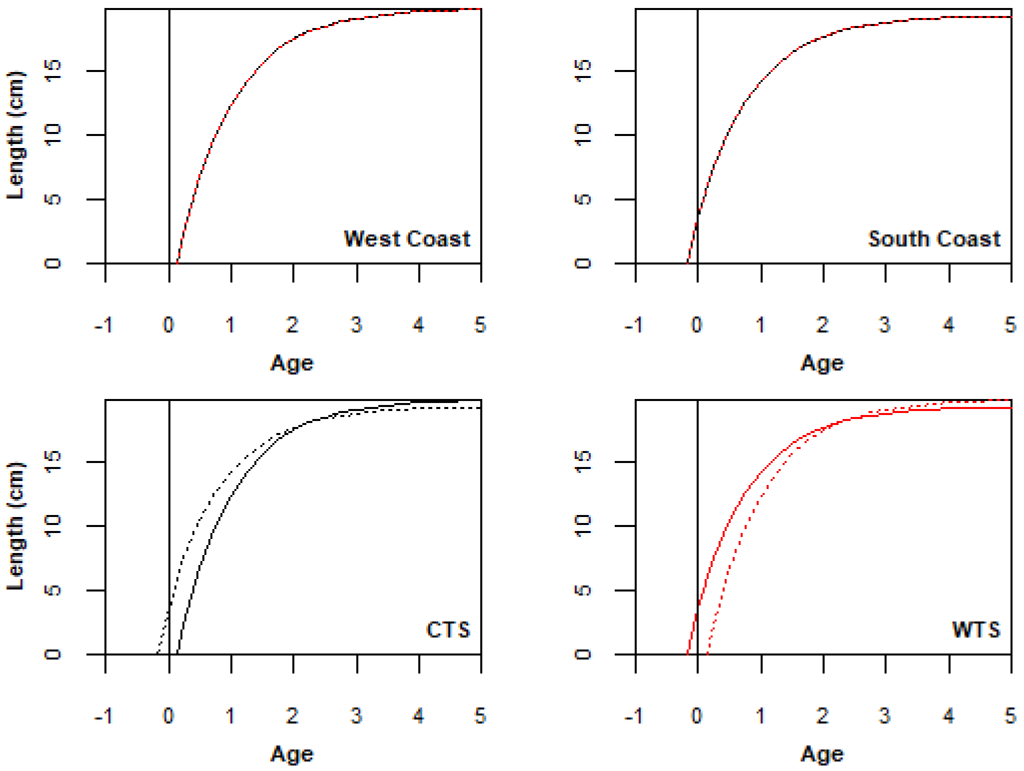


Figure 6c. As for Figure 6a, but for S_{13*}.

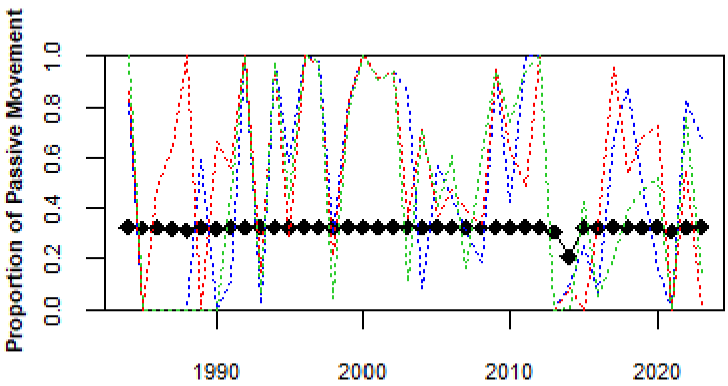


Figure 7. Model estimated proportion of WTS recruits spawned off the South Coast which are passively transported to the West Coast for S_0 (black solid line), S_{14} (blue dotted line), S_{15} (red dotted line) and S_{19} (green dotted line).

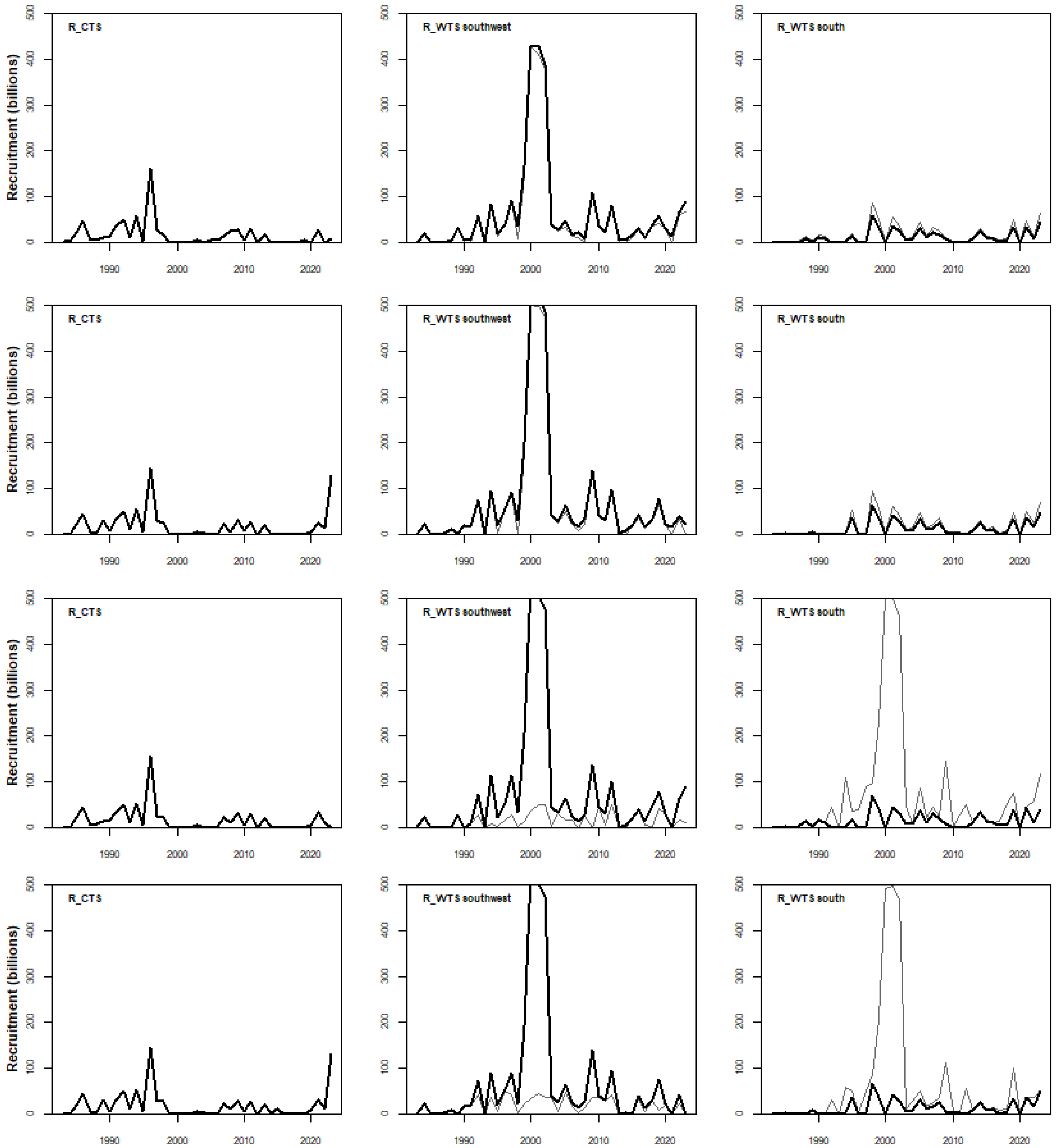


Figure 8a. Model predicted sardine recruitment for S_0 (top row), S_{13} (2nd row), S_{14} (3rd row) and S_{15} (lower row) before (grey lines) and after (black lines) passive movement of WTS from the South to the West Coast.

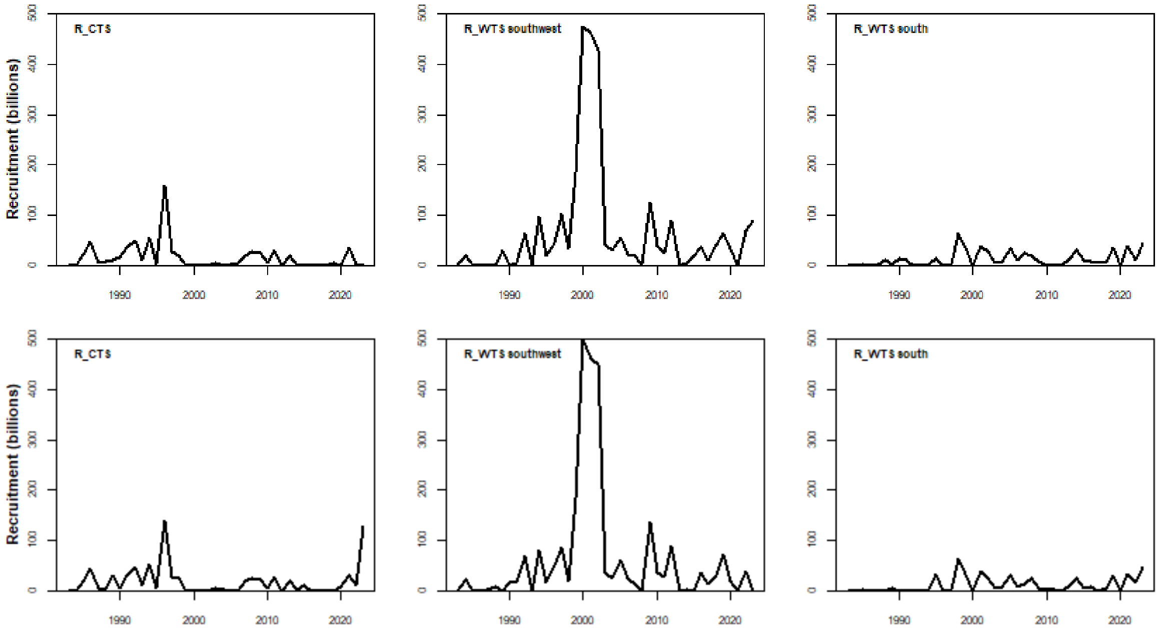


Figure 8b. Model predicted sardine recruitment for S_{16} (top row) and S_{17} (lower row) after WTS is split by coast.

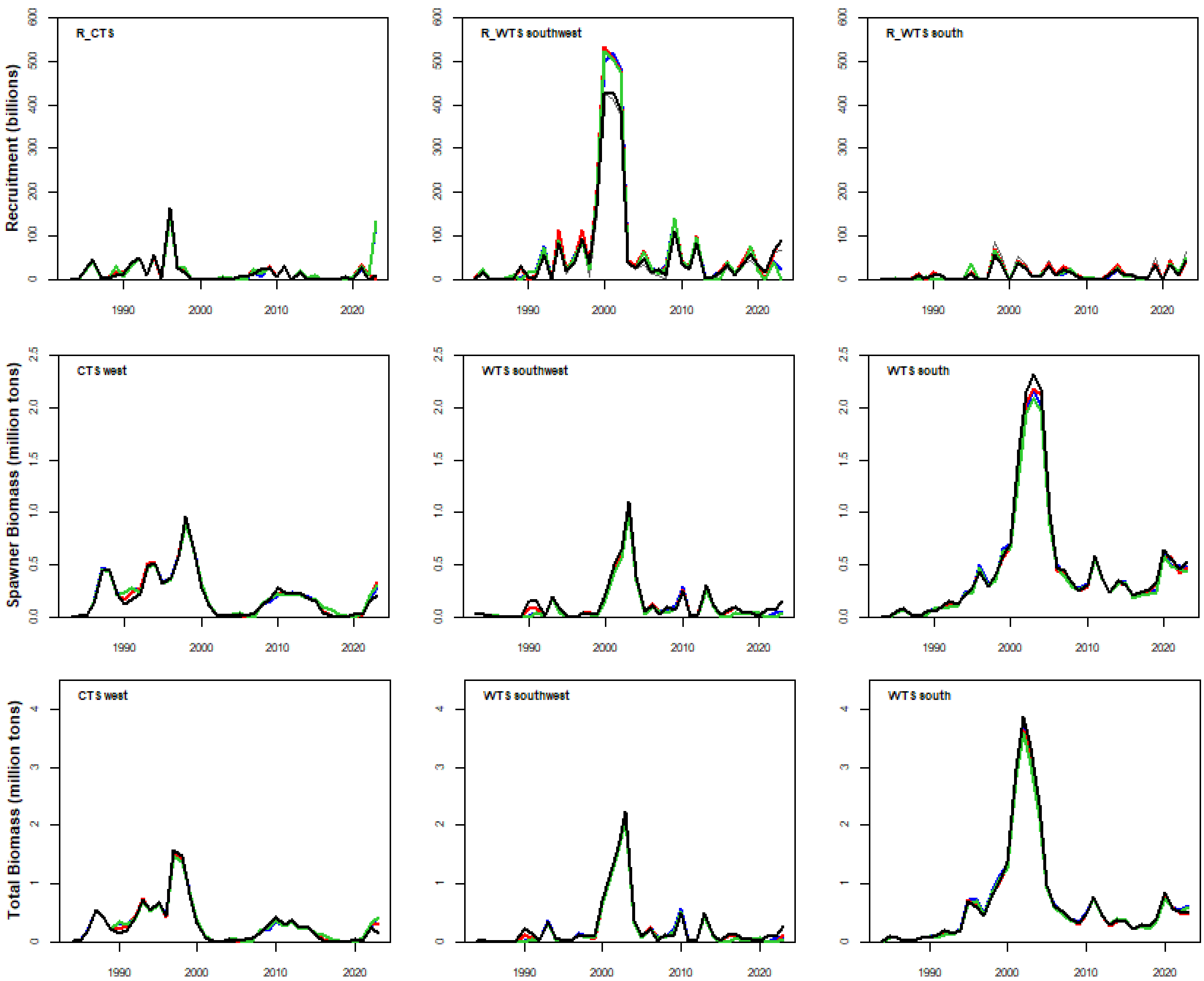


Figure 9a. Model predicted sardine recruitment (top row), spawner biomass (2[nd] row) and total biomass (lower row) from November 1983 to November 2023 for S_0 (black), S_{13} (blue), S_{14} (red) and S_{15} (green).

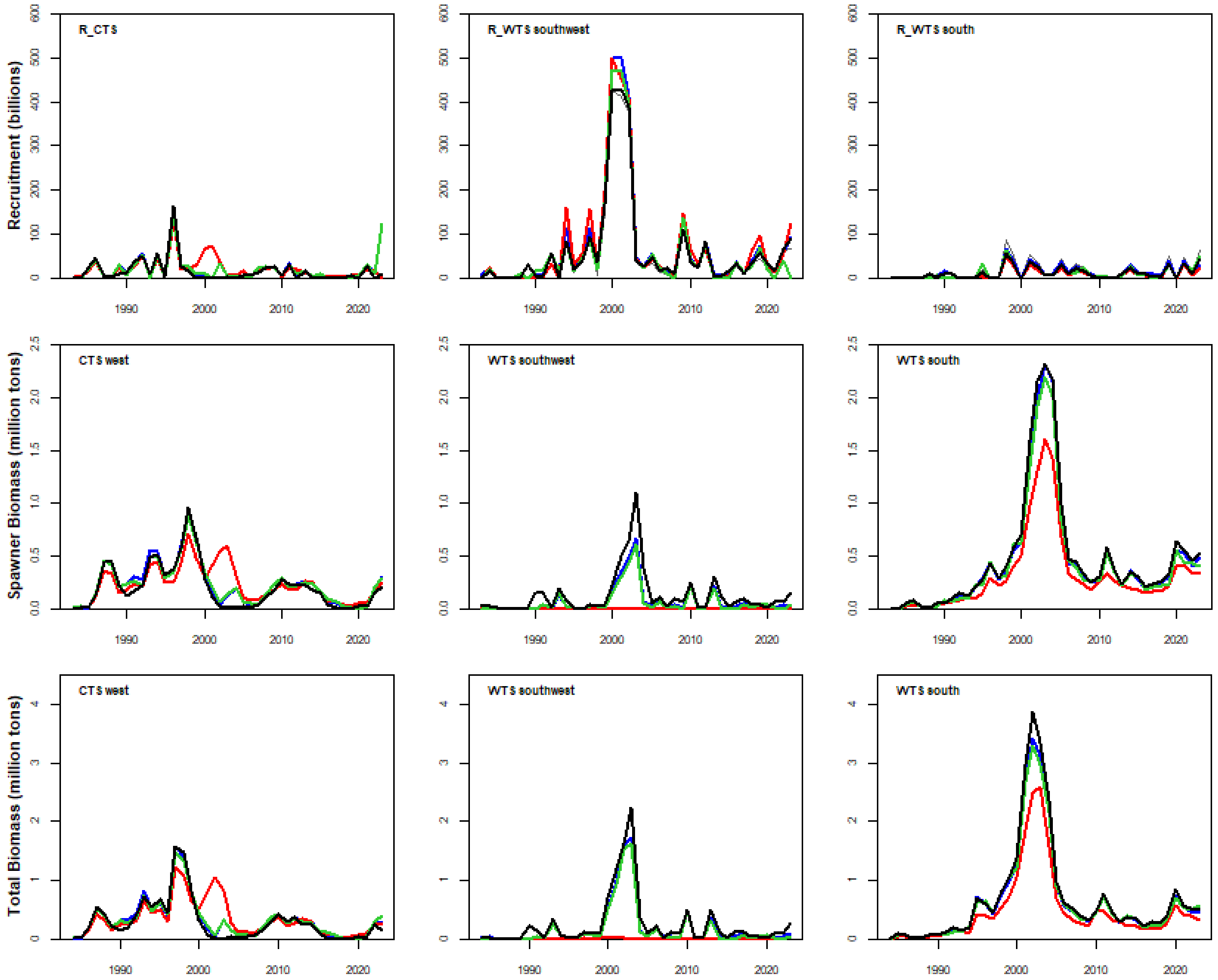
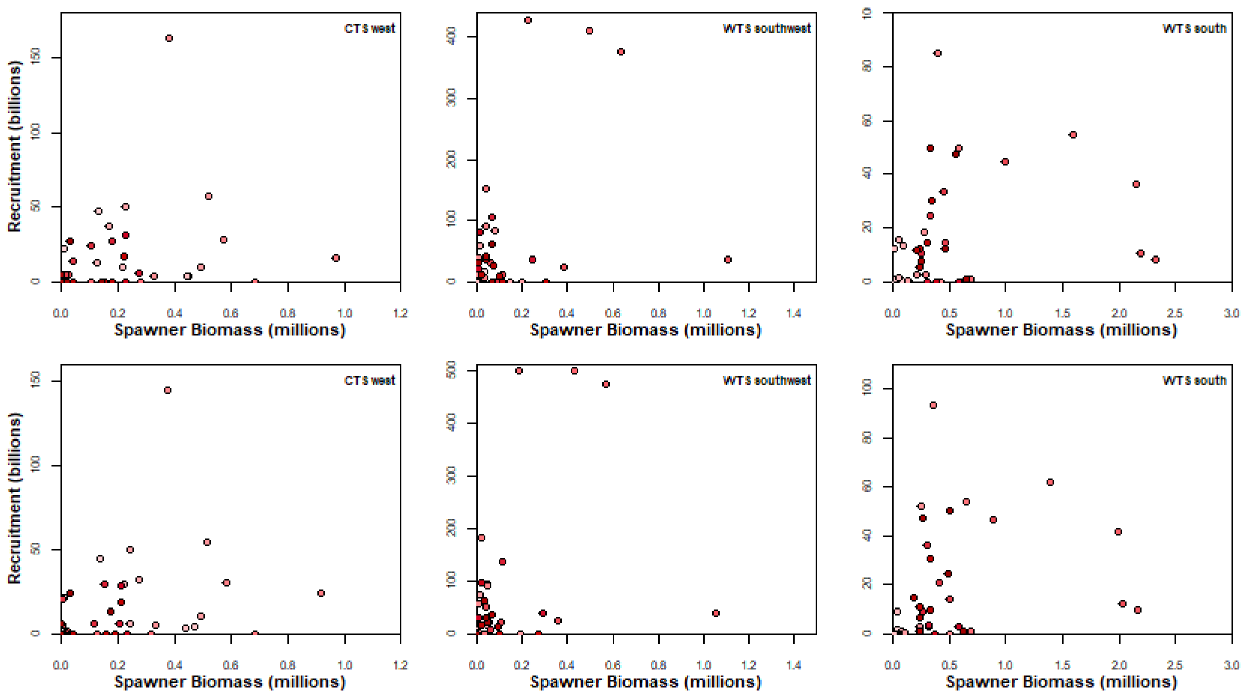


Figure 9b. As for Figure 9a, but for S_0 (black solid line), S_{20} (blue dotted line), S_{21} (red dashed line) and S_{22} (green dashed line).



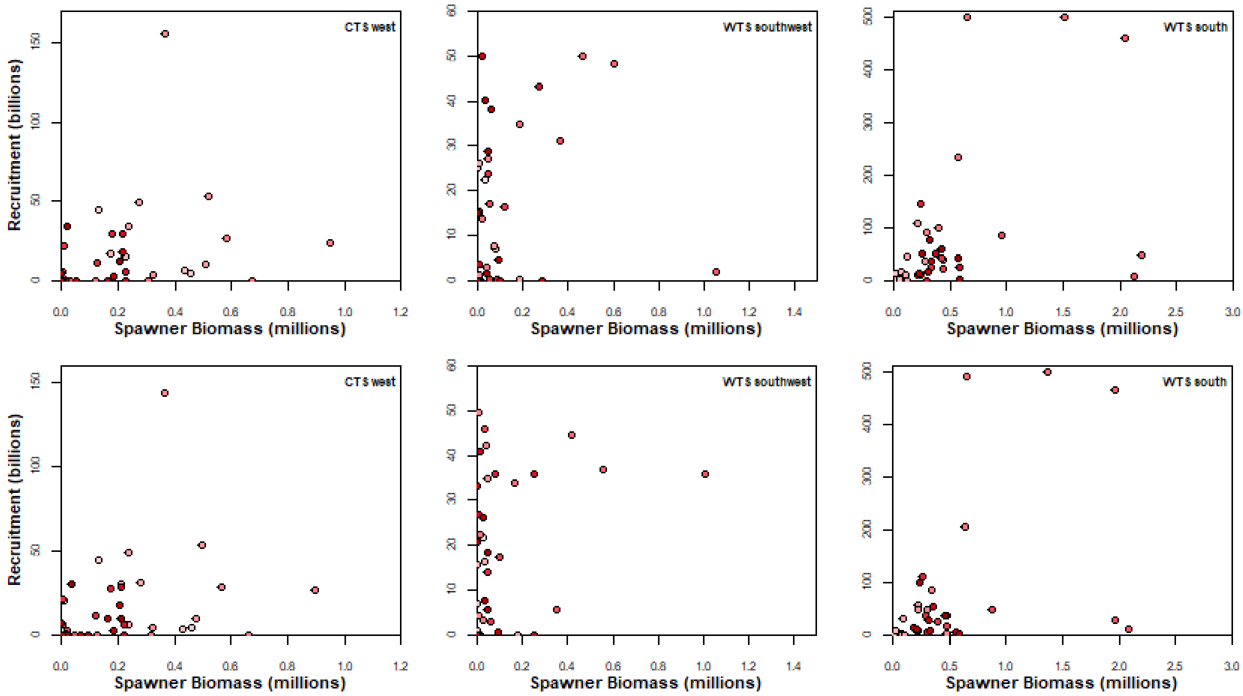


Figure 10a. Model predicted sardine recruitment plotted against spawner biomass, with colouring indicating earlier (lighter) to more recent (darker) years for S_0 (top row), S_{13} (2nd row), S_{14} (3rd row) and S_{15} (lower row).

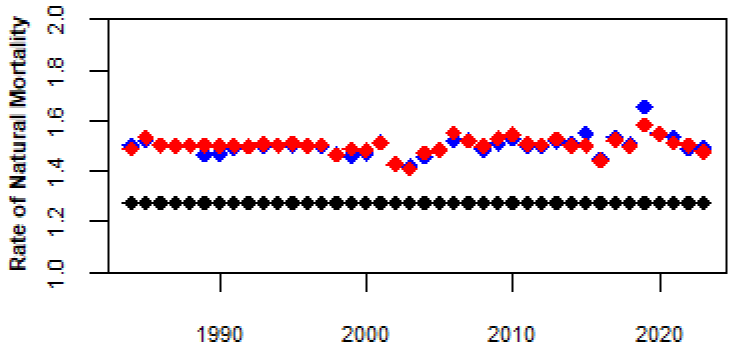


Figure 11. The annual rate of juvenile natural mortality for WTS off the west coast for S_0 (black), S_{18} (blue) and S_{19} (red).